INTRODUCTION TO DEEP LEARNING

RECAP DAY 1

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Schedule

Date	Time	Activity
28.05.2026 Day 1	09:00 - 10:00	Introduction and basics
	10:00 - 10:30	Hands-on I
	10:30 - 10:45	Coffee break
	10:45 - 11:15	Hands-on II
	11:15 - 12:00	Hands-on III (Group Work)
	12:00 - 13:00	Lunch break
	13:00 - 14:00	Advanced concept and Convolutional Neural Network
	14:00 - 14:45	Hands-on IV
	14:45 - 15:15	Recap Day 1
29.05.2026 Day 2	09:00 - 10:00	Deep Generative Models
	10:00 - 10:45	Hands-on III
	10:45 - 11:15	Latest applications, code and knowledge sources
	11:15 - 11:30	Closing

We’re here!

Neural network concepts

Building the network
- Input layer
- Hidden layers
- Output layer

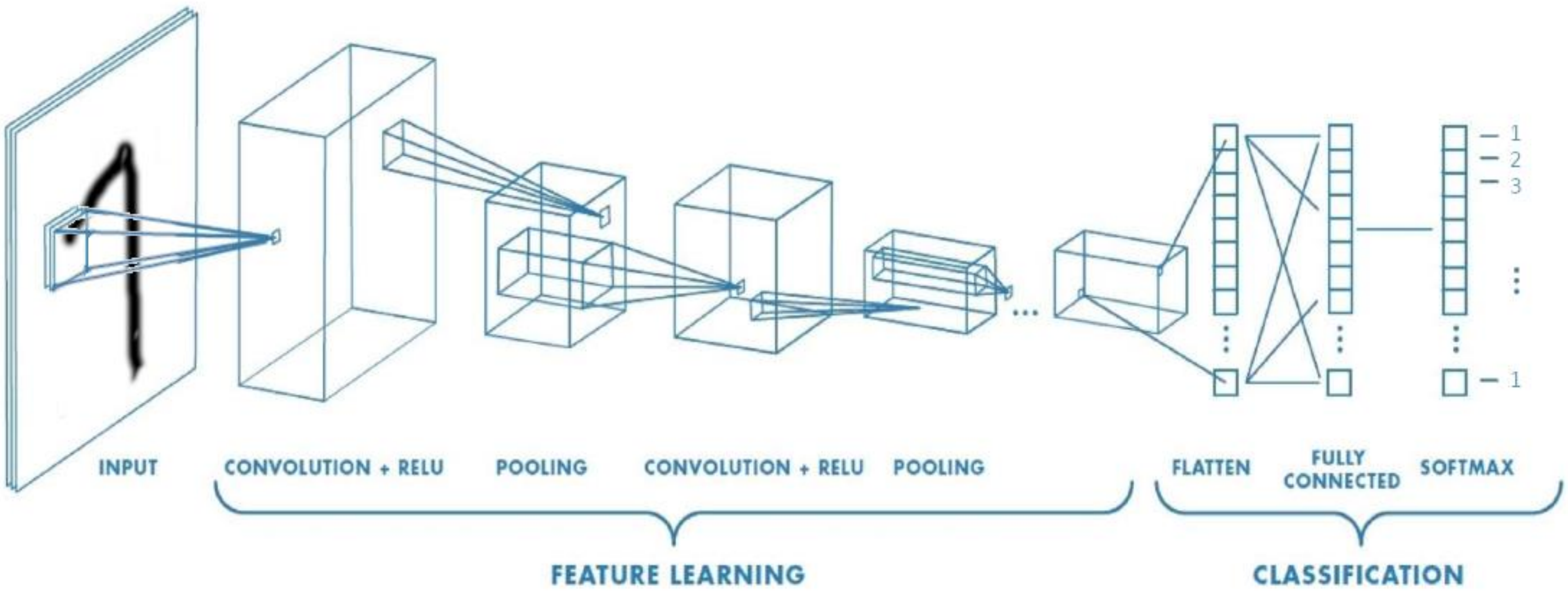
Forward propagate input
- Propagate input through neurons
- Apply non-linearities

Evaluate performance
- Compute the loss of the prediction

Back propagate
- Update weights according to the optimization algorithm

Improving the network
- Regularization,
- Advanced architectures

Convolutional Neural Network



- CONV and POOL layers output high-level features of input
- Fully connected layer uses these features for classifying input image
- Express output as **probability** of image belonging to a particular class

$$\mathrm{softmax}(y_i) = \frac{e^{y_i}}{\sum_j e^{y_j}}$$

Imprint

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